

Sum of iid random variables

Given are

independent identically distributed (continuous) random variables

all with the same distribution. We consider their sum

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Tasks

1. [4 test cases] Compute the mean
- 2.
3. [4 test cases] Compute the variance
- 4.
5. [12 test cases] Given numbers
6. , approximate the probability
7. .

In order to solve tasks 1., 2. and 3., in file `solution.py` implement functions `mean`, `variance` and `probability`, respectively. The distribution of (each of) the random variables

is given as an instance `distribution` of a distribution from `scipy.stats`. (It is one of `norm`, `beta` or `uniform`).

Don't worry, you don't need to know much about the used distributions. Also you do not need to run a simulation. Simply make use of functions that are offered by the `distribution` object already, such as

- `pdf(x)`: probability density function at
-
- `cdf(x)`: cummulative distribution funtion at
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- `mean()`: mean of the given distribution
- `std()`: standard deviation of the given distribution
- `var()`: variance of the distribution

Make use of the Central Limit Theorem for 3. You can assume that

is large enough such that the approximation is good. A random variable that is normally distributed with mean

and standard deviation

can, for example, be instantiated using `normal = scipy.stats.norm(m, s)`

Input / Output examples

Expected at the command line are

- the distribution (one of `beta`, `uniform` or `normal`)
- two shape parameters
- and
- for the distribution
- the number
- of random variables in the sum
- a lower and upper bound for
-

Example Input

```
Enter <dist> <a> <b> <n> <low> <high>:
uniform 0 1 30 10 20
```

Example Output

```
E(Sn) = 15.0
Var(Sn) = 2.5
P(Sn in [ 10.0 , 20.0 ]) = 0.9984 ...
```

When testing or submitting solutions, a range of valid results is part of the input additionally (not shown here).

Always submit partial solutions

Do not hardcode, we will exchange the test values.